

Research Statement

I am currently a master's student in mathematics at Sun Yat-sen University. My primary mathematical interest lies in algebraic geometry, with a current focus on the syzygies of linear series. I am fascinated by how geometric shapes are connected with abstract algebra, especially commutative algebra in algebraic geometry. I hope to further study topics and problems in algebraic geometry, which is the fundamental reason why I want to pursue a doctorate in mathematics. I also find great joy in discovering problems and doing creative work in my research, which motivates me to continue on my academic path. My long-term goal is to contribute original research to the field and to mentor students who share my passion for mathematics.

During my undergraduate studies, I majored in mathematics at a large public university in China, where I received systematic and comprehensive training in the subject. I took specialized courses in analysis, algebra, geometry, and topology and performed well in them, which built a solid mathematical foundation for me.

Because of my outstanding academic record and overall performance, I was recommended for admission to the master's program without taking the entrance examination. During my graduate studies, I completed advanced courses such as Riemannian Geometry, Algebraic Topology, and Commutative Algebra with excellent results, which strengthened my foundation in pure mathematics.

My research experience began with my undergraduate thesis, "Minkowski Sum of Lattice Points in Lattice Polytopes." In the language of algebraic geometry, this problem asks whether a projective toric variety embedded by an ample line bundle is projectively normal, where the toric variety and the line bundle are derived from a lattice polytope. This project introduced me to algebraic geometry and toric varieties, and I was attracted by how they transform complicated geometric structures into accessible combinatorial data derived from polytopes and fans, which serves as a dictionary between algebraic geometry and combinatorics.

In my master's program, I continued in this direction, studying the positivity of linear series and syzygies on toric varieties. The concept of syzygies originally described the shape of free resolutions and Betti numbers in commutative algebra. From the 20th century onward, syzygies connected the structure of minimal free resolutions to the geometry of a projective variety embedded in projective space, such as its projective normality, generation by quadrics, and higher-order relations. Later in the 1980s, M. Green introduced Koszul cohomology and defined Property N_p to describe syzygies.

My work led to the paper "On covering simplices by dilations in dimensions 3 and 4" on arXiv. In this article, I developed a "covering by dilations" method to prove Property N_0 for a line bundle on toric varieties (equivalently, for their associated lattice polytopes). Using this

method, I established Property N_0 for a family of simplicial toric varieties with large lattice length in dimensions 3 and 4. Throughout this project, I combined theoretical arguments with computational experiments using Python and Macaulay2.

My research experiences have strengthened both my technical skills and my research maturity: I learned the fundamentals of algebraic geometry, found real joy in exploring problems, and worked across related topics including geometric invariant theory, stability of vector bundles, moduli spaces, and syzygies and resolutions. More importantly, these experiences have prepared me for doctoral study by teaching me to navigate both abstract theory and concrete computation, approach problems from multiple perspectives, and stay curious amid uncertainty while learning through trial and error.

Furthermore, I gave a series of talks in small reading seminars on Introduction to Toric Varieties and Graded Syzygies at my university. Serving as a speaker allowed me to gain deeper insights through discussions and enhanced my communication skills, particularly in learning how to focus on the core of a problem and convey complex ideas clearly. For my strong academic performance, I received the Graduate Scholarship from my university for three consecutive years.

Beyond coursework, I have also actively participated in various academic activities to communicate with algebraic geometers from all over the world and to appreciate different flavors of algebraic geometry. For instance, I learned about multiplier ideals, V-filtrations, and K-stability theory of singularities during the Algebraic Geometry Summer School held in Shanghai in 2024, as well as combinatorial aspects of algebraic geometry in 2025 SRI in Algebraic Geometry held at Colorado State University.

Teaching is also an important part of my development. As a teaching assistant for undergraduate linear algebra courses, I assisted instructors and tutored students, solving student's problem and building a supportive learning environment. I believe that teaching is a crucial way to refine one's own understanding of mathematical concepts and support the next generation of mathematicians.

In conclusion, my academic path has been guided by curiosity about structures in algebraic geometry and by persistence in tackling challenging problems. Through my research, I have gradually developed a strong interest in the interface between algebraic geometry and combinatorics. An exciting theme in this area is the use of discrete and combinatorial tools to understand and clarify complex geometric structures, as in the toric geometry. In the future, I hope to further broaden my understanding of the field, strengthen my ability to conduct independent research, and contribute to the frontiers of algebraic geometry through original work.